



The RACE For Speed

For years, Intel and AMD upped speeds the gigahertz way. They're now pushing dual-core processors. We look at the why and the what of dual-core

Ram Mohan Rao

The only constant is change. In the computer world, the only constant is increase in speeds—read gigahertz. Ever since the public got sold on the idea of faster processors, Intel and AMD and the smaller processor manufacturers have been making their millions off just one idea—that people want faster computers. Of course, many things go into the making of a faster computer, but a faster processor is perhaps most important. So how do AMD and Intel make processors faster?

In what follows, we dwell on the *why* and *what else* and *what if* of dual-core processors more than on the processors themselves. This is because the actual concept is straightforward enough—it's the peripheral issues that are more pressing.

So, how does one make a faster processor? The first and most obvious answer is to make changes in the design of the processor such that it is intrinsically faster. To illustrate, pipeline implementations (see box *Instructions And Pipelines*) have a huge impact on processor performance—and breakthroughs that improve processor design come about once in a while. But some fundamental barriers are being reached, and the need is for a quick-fix solution to the 'gigahertz problem': faster processors, right now!

The second thing one could do is increase clock speeds. But that's another fundamental barrier that's being reached: the rate at which chips have been shrinking, manufacturers can't seem to achieve higher clock speeds without overheating the chip. Mandatory water cooling? That's a possibility... But even then, the heat we're talking about is due to the power that's supplied to the chip, and it takes a lot of power to get a